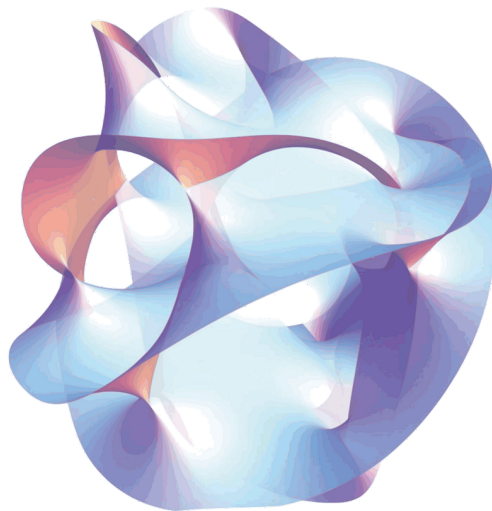


Dimensional reduction in supergravity and duality symmetries

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Notations and abbreviations

For the abbreviations we will use

- GR for general relativity

We will during this internship use the vielbein formalism. We will use the same notation as in [1] which is

- Greek hatted letter for curved indices ($\hat{\mu}, \hat{\nu}, \hat{\rho}, \dots$)
- Latin hatted letter for flat indices ($\hat{p}, \hat{m}, \hat{n}, \dots$)
- For space-time indices taking D values, late greek letters are used in the curved case ($\mu, \nu, \rho, \dots$) and Latin letters in the flat case (p, m, n)
- For internal indices taking E values, early Greek letters in the curved case ($\alpha, \beta, \gamma, \dots$) and Latin letters in the flat case (a,b,c)

The general coordinate $x^{\hat{\mu}}$ will therefore be made of space time coordinate x^{μ} and internal coordinate y^{α} . The flat lorentz

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1 Introduction

Since the 19th-century unification of electricity and magnetism, and the later development of classical and quantum field theory, physicists have been looking for a single mathematical framework that could describe all fundamental interactions at once. A lot of progress has been made for the electromagnetic, weak and strong forces, which are now successfully unified in the Standard Model. Gravity, however, remains the main stumbling block. Unlike the other forces, it resists quantization in a straightforward way, and its description by General Relativity [2] sits completely apart from the quantum field-theoretic language we use for everything else.

One of the earliest attempts to change this was proposed by Kaluza in 1921 [3] and later refined by Klein: by adding one extra spatial dimension compactified on a small circle, five-dimensional General Relativity naturally reduces to four-dimensional Einstein gravity coupled to electromagnetism. Even if the original theory had problems, this Kaluza-Klein mechanism opened a whole new way of thinking about unification, and it is now at the heart of modern approaches like supergravity [4] and string theory.

2 Theoretical tool

Here you will find some GR reminder, which hopefully will make it easier to understand the mathematical foundation on which this internship was build

2.1 spin connexion, vielbein and flatspace

In general relativity we often encounter horrible metrics such as the Kerr-Newman metric. A good idea we can have is to then work in an orthonormal basis $\{e_\mu^a\}$, which follow:

$$g_{\mu\nu}V_a^\mu V_b^\nu = \eta_{ab} \quad (1)$$

where g is the original metric and η is the Minkowski metric.

2.2 Groupe and gauge transformation

3 Lagrangian field theory

3.1 Lagrangian density

In field theory, just as Newton's second law is "volumized" into the Navier-Stokes equations by distributing quantities over space, we replace discrete degrees of freedom with continuous field densities defined at every point in spacetime. Therefore we will rewrite the Lagrangian as the integral over space of a new quantities called Lagrangian density, which will often referred to as just the Lagrangian. Thus we obtain this equation:

$$L = \iiint \mathcal{L}(\eta_\rho, \frac{\partial \eta_\rho}{\partial x^\nu}) dx dy dz \quad (2)$$

Now using Einstein convention we can rewrite the action in a new light :

$$S = \int \mathcal{L} dx^\mu \quad (3)$$

3.2 General exemple for scalar and vectorial fields, and result for two forms

In field theory, the Lagrangian has a classical prescription for scalar and vector fields, but no general rule exists for other cases each field type requires its own treatment. For this work, only the scalar, vector, and two-form Lagrangians are needed, along with the metric, which is handled by Einstein's equations.

So for a scalar field we have :

$$\mathcal{L} = -\frac{1}{2} \partial_\mu \Phi \partial^\mu \Phi \quad (4)$$

and for a vectorial field we have

$$\mathcal{L} = -\frac{1}{2} F_{\mu\nu} F^{\mu\nu} \quad (5)$$

where for the field A

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

In a vacuum the Einstein equation give us

$$\mathcal{L} = R$$

where R is Ricci scalar finally for the Lagrangian of a two form fields we will look in [5] where it is given by the equation :

$$\mathcal{L} = \frac{1}{12} g^{\hat{\mu}\hat{\mu}'} g^{\nu\nu'} g^{\rho\rho'} H_{\mu\nu\rho} H_{\mu'\nu'\rho'} \quad (6)$$

where for the two form $B_{\nu\rho}$ we have

$$H_{\mu\nu\rho} = \partial_\mu B_{\nu\rho} + \text{cyc.perm}$$

4 Reduction of the NS-NS Action in D+E dimension

In this section we perform a Kaluza-Klein reduction. We consider a universe living in $D + E$ dimensions, with $D = 4$, and we try to determine how an observer confined to the D -dimensional world perceives the extra dimensions.

4.1 Hypothesis

We assume that none of the fields depend on the internal coordinates, i.e. for any field $\partial_\alpha A^{\hat{\mu}} = 0$. This means that the reduction will be done on a E-d torus.

4.2 Rewriting the action

We start from the NS-NS action, which is

$$S = \int dx^D \int \frac{dy^E}{\rho(E)} \sqrt{-\hat{g}} e^{\hat{\phi}} (\hat{R} + \hat{g}^{\hat{\mu}\hat{\nu}} \partial_{\hat{\mu}} \hat{\phi} \partial^{\hat{\nu}} \hat{\phi} - \frac{1}{12} g^{\hat{\mu}\hat{\mu}'} g^{\nu\nu'} g^{\rho\rho'} H_{\mu\nu\rho} H_{\mu'\nu'\rho'}) \quad (7)$$

where $\rho(E)$ is a normalisation factor introduced in (7), corresponding in practice to the invariant volume of the internal space.

And where $\hat{R}$ is similar as the classical Hilbert-Einstein action but in $D + E$ dimension, $\hat{\phi}$ is a dilaton field and $-\frac{1}{12} H_{\mu\nu\rho} H^{\mu\nu\rho}$ is the action of a two form. Finally as no field depends on internal components and knowing what $\rho(E)$ we can just simplify this integral. Now we will separate this action into 3 parts one for each action so we have

$$S_{\hat{g}} = \int dx^D \sqrt{-\hat{g}} e^{-\hat{\phi}} \hat{R} \quad (8a)$$

$$S_{\hat{\phi}} = \int dx^D \sqrt{-\hat{g}} e^{-\hat{\phi}} \hat{g}^{\hat{\mu}\hat{\nu}} \partial_{\hat{\mu}} \hat{\phi} \partial^{\hat{\nu}} \hat{\phi} \quad (8b)$$

$$S_{\hat{\phi}} = \int dx^D \sqrt{-\hat{g}} e^{-\hat{\phi}} - \frac{1}{12} H_{\mu\nu\rho} H^{\mu\nu\rho} \quad (8c)$$

Now using the fact that $\hat{R}$ can be written as

$$\hat{R} = \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \hat{R}_{\hat{\mu}\hat{\nu}\hat{r}\hat{s}}$$

where

$$\hat{R}_{\hat{\mu}\hat{\nu}\hat{r}\hat{s}} = \partial_{\hat{\mu}} \hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} + \hat{\omega}_{\hat{\mu}\hat{r}}^{\hat{t}} \hat{\omega}_{\hat{\nu}\hat{t}\hat{s}} - (\mu \leftrightarrow \nu)$$

and the ω are defined by

$$\hat{\omega}_{\hat{t}\hat{r}\hat{s}} = -\hat{\Omega}_{\hat{t}\hat{r}\hat{s}} + \hat{\Omega}_{\hat{r}\hat{s}\hat{t}} - \hat{\Omega}_{\hat{s}\hat{t}\hat{r}}$$

and finally the Ω are defined thanks to the vielbein with

$$\hat{\Omega}_{\hat{r}\hat{s}\hat{t}} = \frac{1}{2} (\hat{V}_{\hat{r}}^{\hat{\mu}} \hat{V}_{\hat{s}}^{\hat{\nu}} - \hat{V}_{\hat{r}}^{\hat{\nu}} \hat{V}_{\hat{s}}^{\hat{\mu}}) \partial_{\hat{\nu}} \hat{V}_{\hat{t}\hat{r}}$$

with all of that we are going to rewrite the action $S_{\hat{g}}$ in termes of $\hat{\omega}$

$$\begin{aligned}
S_{\hat{g}} &= \int dx^D \hat{V} e^{-\hat{\phi}} \hat{R} \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \hat{R}_{\hat{\mu}\hat{\nu}\hat{r}\hat{s}} \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} (\partial_{\hat{\mu}} \hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} + \hat{\omega}_{\hat{\mu}\hat{r}}^{\hat{t}} \hat{\omega}_{\hat{\nu}\hat{t}\hat{s}} - \partial_{\hat{\nu}} \hat{\omega}_{\hat{\mu}\hat{r}\hat{s}} - \hat{\omega}_{\hat{\nu}\hat{r}}^{\hat{t}} \hat{\omega}_{\hat{\mu}\hat{t}\hat{s}}) \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} ([\partial_{\hat{\mu}} \hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} - \partial_{\hat{\nu}} \hat{\omega}_{\hat{\mu}\hat{r}\hat{s}}] + [\hat{\omega}_{\hat{\mu}\hat{r}}^{\hat{t}} \hat{\omega}_{\hat{\nu}\hat{t}\hat{s}} - \hat{\omega}_{\hat{\nu}\hat{r}}^{\hat{t}} \hat{\omega}_{\hat{\mu}\hat{t}\hat{s}}]) \\
&= \int dx^D \{2\hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \partial_{\hat{\mu}} \hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} + \hat{V} e^{-\hat{\phi}} [\hat{\omega}_{\hat{r}}^{\hat{t}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{s}} - \hat{\omega}_{\hat{r}}^{\hat{s}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{t}}]\} \\
&= \int dx^D \{\hat{V} e^{-\hat{\phi}} [\hat{\omega}_{\hat{r}}^{\hat{t}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{s}} - \hat{\omega}_{\hat{r}}^{\hat{s}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{t}}] + \underbrace{\partial_{\hat{\mu}} (2\hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \hat{\omega}_{\hat{\nu}\hat{r}\hat{s}})}_{\text{Border terme which goes to zero}} - 2\hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} \partial_{\hat{\mu}} (\hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}})\} \\
&= \int dx^D \{\hat{V} e^{-\hat{\phi}} [\hat{\omega}_{\hat{r}}^{\hat{t}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{s}} - \hat{\omega}_{\hat{r}}^{\hat{s}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{t}}] - 2\hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} \partial_{\hat{\mu}} (\hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}})\} \\
&= \int dx^D \{\hat{V} e^{-\hat{\phi}} [\hat{\omega}_{\hat{r}}^{\hat{t}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{s}} - \hat{\omega}_{\hat{r}}^{\hat{s}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{t}}] - 2\hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} [\hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \partial_{\hat{\mu}} \hat{V}^{\hat{\nu}\hat{s}} + \hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\nu}\hat{s}} \partial_{\hat{\mu}} \hat{V}^{\hat{\mu}\hat{r}} + e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \partial_{\hat{\mu}} \hat{V} + \hat{V} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \partial_{\hat{\mu}} e^{-\hat{\phi}}]\} \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \{[\hat{\omega}_{\hat{r}}^{\hat{t}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{s}} - \hat{\omega}_{\hat{r}}^{\hat{s}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{t}}] - 2\hat{\omega}_{\hat{\nu}}^{\hat{r}\hat{s}} [\hat{V}^{\hat{\mu}} \hat{V}_s^{\hat{\nu}} V_t^{\hat{\lambda}} \partial_{\hat{\mu}} \hat{V}_{\hat{\lambda}}^{\hat{t}} - \hat{V}_{\hat{r}}^{\hat{\mu}} \hat{V}_t^{\hat{\nu}} \hat{V}_s^{\hat{\lambda}} \partial_{\hat{\mu}} \hat{V}_{\hat{\lambda}}^{\hat{t}} - \hat{V}_s^{\hat{\nu}} \hat{V}_t^{\hat{\mu}} \hat{V}_{\hat{r}}^{\hat{\lambda}} \partial_{\hat{\mu}} \hat{V}_{\hat{\lambda}}^{\hat{t}} - \hat{V}_{\hat{r}}^{\hat{\mu}} \hat{V}_s^{\hat{\nu}} \partial_{\hat{\mu}} \hat{\phi}]\} \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \{[\hat{\omega}_{\hat{r}}^{\hat{t}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{s}} - \hat{\omega}_{\hat{r}}^{\hat{s}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{t}}] - 2\hat{\omega}_s^{\hat{r}\hat{s}} [\hat{V}_{\hat{r}}^{\hat{\mu}} \hat{V}_t^{\hat{\nu}} \partial_{\hat{\mu}} \hat{V}_{\hat{\lambda}}^{\hat{t}} - \hat{V}_t^{\hat{\mu}} \hat{V}_{\hat{r}}^{\hat{\nu}} \partial_{\hat{\mu}} \hat{V}_{\hat{\lambda}}^{\hat{t}} - \hat{V}_{\hat{r}}^{\hat{\mu}} \partial_{\hat{\mu}} \hat{\phi}] + 2\hat{\omega}_t^{\hat{r}\hat{s}} \hat{V}_{\hat{r}}^{\hat{\mu}} \hat{V}_s^{\hat{\nu}} \partial_{\hat{\mu}} \hat{V}_{\hat{\lambda}}^{\hat{t}}\} \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \{[\hat{\omega}_{\hat{r}}^{\hat{t}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{s}} - \hat{\omega}_{\hat{r}}^{\hat{s}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{t}}] - 2\hat{\omega}_s^{\hat{r}\hat{s}} [2\hat{\Omega}_{\hat{t}\hat{r}}^{\hat{t}} - \hat{V}_{\hat{r}}^{\hat{\mu}} \partial_{\hat{\mu}} \hat{\phi}] + 2\hat{\omega}_t^{\hat{r}\hat{s}} \hat{V}_{\hat{r}}^{\hat{\mu}} \hat{V}_s^{\hat{\nu}} \partial_{\hat{\mu}} \hat{V}_{\hat{\lambda}}^{\hat{t}}\} \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \{[\hat{\omega}_{\hat{r}}^{\hat{t}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{s}} - \hat{\omega}_{\hat{r}}^{\hat{s}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{t}}] + 2\hat{\omega}_s^{\hat{r}\hat{s}} [2\hat{\omega}_{\hat{t}\hat{r}}^{\hat{t}} + \hat{V}_{\hat{r}}^{\hat{\mu}} \partial_{\hat{\mu}} \hat{\phi}] + 2\hat{\omega}_t^{\hat{r}\hat{s}} \hat{\Omega}_{\hat{r}\hat{s}}^{\hat{t}}\} \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \{[\hat{\omega}_{\hat{r}}^{\hat{t}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{s}} - \hat{\omega}_{\hat{r}}^{\hat{s}} \hat{\omega}_{\hat{t}\hat{s}}^{\hat{t}}] + 2\hat{\omega}_s^{\hat{r}\hat{s}} [2\hat{\omega}_{\hat{t}\hat{r}}^{\hat{t}} + \hat{V}_{\hat{r}}^{\hat{\mu}} \partial_{\hat{\mu}} \hat{\phi}] - \hat{\omega}_t^{\hat{r}\hat{s}} (\omega_s^{\hat{t}\hat{r}} + \hat{\omega}_{\hat{r}\hat{s}}^{\hat{t}})\} \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \{\hat{\omega}_s^{\hat{t}\hat{r}} \hat{\omega}_{\hat{r}\hat{t}}^{\hat{s}} + \hat{\omega}_{\hat{r}\hat{t}}^{\hat{s}} \hat{\omega}_{\hat{s}\hat{t}}^{\hat{r}} + 2\hat{\omega}_s^{\hat{r}\hat{s}} \hat{V}_{\hat{r}}^{\hat{\mu}} \partial_{\hat{\mu}} \hat{\phi}\}
\end{aligned} \tag{9}$$

4.3 Vielbein Anstaz

Now that we have explicitley written the action in terme of either latin connexion and or the two other fields, what we hate to do is to suppose an anstaz for the vielbein. Why the vielbein and not the metric you may ask? Because as stated in [2.1](#), all the information of the metric can be contained within the vielbein, and has we primarily work with them for the action, it is easier to do an anstaz on them. Now we will do the same anzasts as in the [\[1\]](#), however we will consider that γ they use is null since the begining, and we will use the the same parameter number as in [\[5\]](#). Finally thank to the local lorentz invariance we can choose a triangular parentrisation giving us

$$\hat{V}_{\hat{\mu}}^{\hat{r}} = \begin{pmatrix} V_{\hat{\mu}}^r & A_{\hat{\mu}}^{(1)\beta} \Phi_{\beta}^a \\ 0 & \Phi_{\alpha}^a \end{pmatrix} \tag{10}$$

Now still using what we learned in part [2.1](#) we can compute the inverse vielbein and then the metric and then the inverse metric that gave us

$$\hat{V}_{\hat{\mu}}^{\hat{r}} = \begin{pmatrix} V_{\hat{\mu}}^r & -\delta^{-\gamma} A_{\hat{\mu}}^{\alpha} \\ 0 & \Phi_{\alpha}^a \end{pmatrix}, \tag{11}$$

for the inverse vielbein, and

$$\hat{g}_{\hat{\mu}\hat{\nu}} = \begin{pmatrix} g_{\mu\nu} - A_{\mu}^{\gamma} A_{\nu\lambda} & -A_{\mu\beta} \\ A_{\nu\alpha} & -h_{\alpha\beta} \end{pmatrix}, \text{ and } \hat{g}^{\hat{\mu}\hat{\nu}} = \begin{pmatrix} g^{\mu\nu} & -A^{\mu\beta} \\ -A^{\nu\alpha} & -h^{\alpha\beta} + A_{\rho}^{\beta} A^{\rho\alpha} \end{pmatrix}, \tag{12}$$

for the metric

Now that we have made this anstaz an intersting things to do is to verify what does object are by using the classical transformation $x'^{\hat{\mu}} = x^{\hat{\mu}} + \xi^{\hat{\mu}}(x)$ and then using what we saw in part that with the Lie derivate we can rewrote this transformation as being Using the fact that the theory is invariant under the classical transformation $x'^{\hat{\mu}} = x^{\hat{\mu}} + \xi^{\hat{\mu}}(x)$ [\[1\]](#), which can be rewritten as

$$\delta \hat{V}_{\hat{\mu}}^{\hat{r}} = \xi^{\hat{\rho}} \partial_{\hat{\rho}} \hat{V}_{\hat{\mu}}^{\hat{r}} + \partial_{\hat{\mu}} \xi^{\hat{\rho}} \hat{V}_{\hat{\rho}}^{\hat{r}} \tag{13}$$

we now apply this transformation to each block of the vielbein, in order to see whether it corresponds to some familiar type of field. Let us start with the block $\hat{V}_\mu^r$:

$$\begin{aligned}\delta\hat{V}_\mu^r &= \xi^{\hat{\rho}}\partial_{\hat{\rho}}\hat{V}_\mu^r + \partial_\mu\xi^{\hat{\rho}}\hat{V}_{\hat{\rho}}^r \\ &= \xi^{\hat{\rho}}\partial_{\hat{\rho}}\hat{V}_\mu^r + \partial_\mu\xi^\nu\hat{V}_\nu^r + \underbrace{\partial_\mu\xi^\alpha\hat{V}_\alpha^r}_{\text{this is null}} \\ &= \xi^{\hat{\rho}}\partial_{\hat{\rho}}(\delta^\gamma V_\mu^r + \delta^\gamma\partial_\mu\xi^\nu V_\nu^r) \\ &= \delta^\gamma(\xi^{\hat{\rho}}\partial_{\hat{\rho}}V_\mu^r + \partial_\mu\xi^\nu V_\nu^r) + \xi^{\hat{\rho}}\partial_{\hat{\rho}}\delta^\gamma V_\mu^r\end{aligned}$$

We can also redo this computation using a different method:

$$\begin{aligned}\delta\hat{V}_\mu^r &= \delta(\delta^\gamma V_\mu^r) \\ &= \delta(\delta^\gamma)V_\mu^r + \delta^\gamma\delta(V_\mu^r)\end{aligned}$$

and by identification we find

$$\delta V_\mu^r = \xi^{\hat{\rho}}\partial_{\hat{\rho}}V_\mu^r + \partial_\mu\xi^\nu V_\nu^r$$

which is exactly the transformation law of a vector field under a Lie derivative [4]. Applying the same method to the other blocks, we obtain, for Φ_α^a :

$$\delta\Phi_\alpha^a = \xi^\rho\partial_\rho\Phi_\alpha^a$$

which is exactly the transformation law of a scalar field. Finally, for A_μ^α we obtain:

$$A_\mu^\alpha = \xi^\rho\partial_\rho A_\mu^\alpha + \partial_\mu\xi^\rho A_\rho^\alpha + \frac{\partial_\mu\xi^\alpha}{2\kappa}$$

which corresponds to the transformation law of a vector field, up to an extra correction term. One may further note that, restricting to ξ^α only, we obtain

$$\delta V_\mu^r = \delta\Phi_\alpha^a = 0, \quad (14a)$$

$$\delta A_\mu^\alpha = \frac{\partial_\mu\xi^\alpha}{2\kappa}. \quad (14b)$$

so that what remains of the D -dimensional diffeomorphisms, once restricted to ξ^α independent of the internal coordinates y , corresponds to a set of E abelian gauge symmetries. So now we can precisely describe the component of the vielbein within this expression, $\delta = \det \Phi$. The vielbein in (??) thus contains the D -dimensional vielbein (i.e. the graviton), E vector fields A_μ^α , and E^2 scalar fields Φ_α^a . However, because of the local Lorentz invariance in the internal directions, only $\frac{1}{2}E(E+1)$ of these scalars actually correspond to propagating degrees of freedom.

4.4 Ricci scalar reduction

Now that we better know the tool we will use, we can begin the calculation. For a starter we will do the Einstein action in $D+E$ dimension twisted by the addition of the dilation.

Firstly let's begin by expanding completely the $S_{\hat{g}}$, this will show us that we technically have to compute 8 ω , however thanks to the antisymmetry of the last to index we can reduce that to 6. So the action is

$$\begin{aligned}S = -\frac{1}{4\kappa^2} \int d^D x \int \frac{d^E y}{\rho(E)} \hat{V} \Big[&\hat{\omega}_{bc}^a \hat{\omega}^{bc}{}_a - \hat{\omega}_{tb}^a \hat{\omega}^{bt}{}_a + \hat{\omega}_{sc}^a \hat{\omega}^{sc}{}_a + \hat{\omega}_{st}^a \hat{\omega}^{st}{}_a - \hat{\omega}_{bc}^r \hat{\omega}^{bc}{}_r - \hat{\omega}_{tb}^r \hat{\omega}^{bt}{}_r - \hat{\omega}_{sc}^r \hat{\omega}^{sc}{}_r \\ &\hat{\omega}_{st}^r \hat{\omega}^{st}{}_r + \hat{\omega}_{ac}^a \hat{\omega}^{bc}{}_b + \hat{\omega}_{ta}^a \hat{\omega}^{bt}{}_b + \hat{\omega}_{ac}^a \hat{\omega}^{sc}{}_c - \hat{\omega}_{ta}^a \hat{\omega}^{st}{}_s + \hat{\omega}_{rc}^r \hat{\omega}^{bc}{}_b - \hat{\omega}_{rt}^r \hat{\omega}^{bt}{}_b + \hat{\omega}_{rc}^r \hat{\omega}^{sc}{}_c + \hat{\omega}_{rt}^r \hat{\omega}^{st}{}_s + \\ &2\hat{\omega}_s{}^{rs} \hat{V}_r^\mu \partial_\mu \hat{\phi} + 2\hat{\omega}_b{}^{rb} \hat{V}_r^\mu \partial_\mu \hat{\phi} \Big] \quad (15)\end{aligned}$$

It then becomes clear that we only need to compute six components of ω , namely

$$\begin{aligned}\hat{\omega}_{pmn}, \hat{\omega}_{pma}, \hat{\omega}_{pab}, \\ \hat{\omega}_{cmn}, \hat{\omega}_{cma}, \hat{\omega}_{cab},\end{aligned}$$

One can also check that computing these six components only requires the following eight components of Ω :

$$\begin{aligned}\hat{\Omega}_{pmn}, \\ \hat{\Omega}_{pma}, \hat{\Omega}_{pam}, \hat{\Omega}_{apm}, \\ \hat{\Omega}_{mab}, \hat{\Omega}_{amb}, \hat{\Omega}_{abm}, \\ \hat{\Omega}_{abc},\end{aligned}$$

Using (??), (??) and (??), we can now set out a clear strategy: first compute $\hat{V}_{\hat{r}}^{\hat{\mu}}$ and $\hat{V}_{\hat{\mu}\hat{r}}$, then the Ω 's, and finally the ω 's. The inverse vielbein is easy to obtain, since

$$\hat{V}_{\hat{\mu}}^{\hat{s}} \hat{V}_{\hat{r}}^{\hat{\mu}} = \delta_{\hat{s}}^{\hat{r}}$$

so that

$$\hat{V}_{\hat{\mu}}^{\hat{r}} = \begin{pmatrix} \delta^{-\gamma} V_r^\mu & -2\kappa \delta^{-\gamma} A_r^\alpha \\ 0 & \Phi_a^\alpha \end{pmatrix} \quad (16)$$

Similarly, for $\hat{V}_{\hat{\mu}\hat{r}}$, using

$$\hat{V}_{\hat{r}}^{\hat{\mu}} \hat{V}_{\hat{\mu}\hat{s}} = \eta_{\hat{r}\hat{s}}$$

we obtain

$$\hat{V}_{\hat{\mu}\hat{s}} = \begin{pmatrix} \delta^\gamma V_{\mu s} & 2\kappa A_\mu^\alpha \Phi_{\alpha a} \\ 0 & \Phi_{\alpha a} \end{pmatrix} \quad (17)$$

Let us now turn to the computation of the Ω 's themselves. Starting with $\hat{\Omega}_{pmn}$:

$$\begin{aligned} \hat{\Omega}_{pmn} &= \frac{1}{2} (\hat{V}_p^{\hat{\mu}} \hat{V}_m^\nu - \hat{V}_m^{\hat{\mu}} \hat{V}_p^\nu) \partial_\nu \hat{V}_{\hat{\mu}n} \\ &= \frac{1}{2} (\hat{V}_p^\mu \hat{V}_m^\nu - \hat{V}_m^\mu \hat{V}_p^\nu) \partial_\nu \hat{V}_{\mu n} \\ &= \frac{1}{2} (\hat{V}_p^\mu \hat{V}_m^\nu - \hat{V}_m^\mu \hat{V}_p^\nu) \partial_\nu \hat{V}_{\mu n} \\ &= \frac{\delta^{-2\gamma}}{2} (V_p^\mu V_m^\nu - V_m^\mu V_p^\nu) \partial_\nu (\delta^\gamma V_{\mu n}) \\ &= \frac{\delta^{-2\gamma}}{2} (V_p^\mu V_m^\nu - V_m^\mu V_p^\nu) (\partial_\nu (\delta^\gamma) V_{\mu n} + \delta^\gamma \partial_\nu (V_{\mu n})) \\ &= \frac{\delta^{-2\gamma}}{2} (V_p^\mu V_m^\nu - V_m^\mu V_p^\nu) (\gamma \partial_\nu (\delta) \delta^{\gamma-1} V_{\mu n} + \delta^\gamma \partial_\nu (V_{\mu n})) \\ &= \frac{\delta^{-\gamma}}{2} (V_p^\mu V_m^\nu - V_m^\mu V_p^\nu) (\gamma \partial_\nu (\delta) \delta^{-1} V_{\mu n} + \partial_\nu (V_{\mu n})) \\ &= \frac{\delta^{-\gamma}}{2} (V_p^\mu V_m^\nu - V_m^\mu V_p^\nu) (\gamma \partial_\nu (\log \delta) V_{\mu n} + \partial_\nu (V_{\mu n})) \end{aligned}$$

For $\hat{\Omega}_{pma}$, we find:

$$\begin{aligned} \hat{\Omega}_{pma} &= \frac{1}{2} (\hat{V}_p^{\hat{\mu}} \hat{V}_m^\nu - \hat{V}_m^{\hat{\mu}} \hat{V}_p^\nu) \partial_\nu \hat{V}_{\hat{\mu}a} \\ &= -\kappa \delta^{-2\gamma} (V_p^\mu V_m^\nu - V_m^\mu V_p^\nu) \partial_\nu (A_\mu^\alpha \Phi_{a\alpha}) + \kappa \delta^{-\gamma} (A_p^\alpha V_m^\nu - \hat{A}_m^\alpha V_p^\nu) \partial_\nu (\Phi_{a\alpha}) \\ &= -\kappa \delta^{-2\gamma} (V_p^\mu V_m^\nu - V_m^\mu V_p^\nu) (\partial_\nu (A_\mu^\alpha) \Phi_{a\alpha} + \partial_\nu (\Phi_{a\alpha}) A_\mu^\alpha) + \kappa \delta^{-\gamma} (A_p^\alpha V_m^\nu - \hat{A}_m^\alpha V_p^\nu) \partial_\nu (\Phi_{a\alpha}) \\ &= -\kappa \delta^{-2\gamma} (V_p^\mu V_m^\nu - V_m^\mu V_p^\nu) (\partial_\nu (A_\mu^\alpha) \Phi_{a\alpha} + \partial_\nu (\Phi_{a\alpha}) A_\mu^\alpha) + (A_p^\alpha V_m^\nu - \hat{A}_m^\alpha V_p^\nu) \partial_\nu (\Phi_{a\alpha}) \\ &= -\kappa \delta^{-2\gamma} (V_p^\mu V_m^\nu - V_m^\mu V_p^\nu) \partial_\nu (A_\mu^\alpha) \Phi_{a\alpha} \\ &= \kappa \delta^{-2\gamma} (F_{pm}^\alpha \Phi_{a\alpha} - \partial_\nu (V_p^\mu V_m^\nu) A_\mu^\alpha \Phi_{a\alpha} + \partial_\nu (V_p^\nu V_m^\mu) A_\mu^\alpha \Phi_{a\alpha}) \\ &= \kappa \delta^{-2\gamma} F_{pm}^\alpha \Phi_{a\alpha} \end{aligned}$$

For $\hat{\Omega}_{pam}$:

$$\begin{aligned} \hat{\Omega}_{pam} &= \frac{1}{2} (\hat{V}_p^{\hat{\mu}} \hat{V}_a^\nu - \hat{V}_a^{\hat{\mu}} \hat{V}_p^\nu) \partial_\nu \hat{V}_{\hat{\mu}m} \\ &= \frac{1}{2} (\hat{V}_p^{\hat{\mu}} \hat{V}_a^\nu - \hat{V}_a^{\hat{\mu}} \hat{V}_p^\nu) \partial_\nu \hat{V}_{\hat{\mu}m} \\ &= 0 \end{aligned}$$

The same result holds for $\hat{\Omega}_{apm}$.

For $\hat{\Omega}_{pab}$:

$$\begin{aligned}
\hat{\Omega}_{pab} &= \frac{1}{2}(\hat{V}_p^{\hat{\mu}}\hat{V}_a^{\nu} - \hat{V}_a^{\hat{\mu}}\hat{V}_p^{\nu})\partial_{\nu}\hat{V}_{\hat{\mu}b} \\
&= \frac{-\delta^{-\gamma}}{2}\hat{V}_a^{\hat{\mu}}V_p^{\nu}\partial_{\nu}\hat{V}_{\hat{\mu}b} \\
&= \frac{\delta^{-\gamma}}{2}\Phi_a^{\alpha}V_p^{\nu}\partial_{\nu}\Phi_{\alpha b} \\
&= \frac{\delta^{-\gamma}}{2}\Phi_a^{\alpha}V_p^{\nu}\partial_{\nu}(h_{\alpha\beta}\Phi_b^{\beta}) \\
&= \frac{\delta^{-\gamma}}{2}\Phi_a^{\alpha}\Phi_b^{\beta}V_p^{\nu}\partial_{\nu}h_{\alpha\beta} + \frac{\delta^{-\gamma}}{2}\Phi_a^{\alpha}V_p^{\nu}h_{\alpha\beta}\partial_{\nu}\Phi_b^{\beta} \\
&= \frac{\delta^{-\gamma}}{2}\Phi_a^{\alpha}\Phi_b^{\beta}V_p^{\nu}\partial_{\nu}h_{\alpha\beta} + \frac{\delta^{-\gamma}}{2}\Phi_{\beta a}V_p^{\nu}\partial_{\nu}\Phi_b^{\beta} \\
&= \frac{\delta^{-\gamma}}{2}\Phi_a^{\alpha}\Phi_b^{\beta}V_p^{\nu}\partial_{\nu}h_{\alpha\beta} + \frac{\delta^{-\gamma}}{2}V_p^{\nu}\partial_{\nu}(\underbrace{\Phi_{\beta a}\Phi_b^{\beta}}_{=\delta_{ab} \text{ i.e. is constant}}) - \Phi_b^{\beta}V_p^{\nu}\partial_{\nu}\Phi_{\beta a} \\
&= \frac{\delta^{-\gamma}}{2}\Phi_a^{\alpha}\Phi_b^{\beta}V_p^{\nu}\partial_{\nu}h_{\alpha\beta} - \Phi_b^{\beta}V_p^{\nu}\partial_{\nu}\Phi_{\beta a}
\end{aligned}$$

For $\hat{\Omega}_{apb}$:

$$\begin{aligned}
\hat{\Omega}_{apb} &= \frac{1}{2}(\hat{V}_a^{\hat{\mu}}\hat{V}_p^{\nu} - \hat{V}_p^{\hat{\mu}}\hat{V}_a^{\nu})\partial_{\nu}\hat{V}_{\hat{\mu}b} \\
&= \frac{1}{2}\hat{V}_a^{\hat{\mu}}\hat{V}_p^{\nu}\partial_{\nu}\hat{V}_{\hat{\mu}b} \\
&= \frac{1}{2}\hat{V}_a^{\alpha}\hat{V}_p^{\nu}\partial_{\nu}\hat{V}_{\alpha b} \\
&= \frac{-\delta^{-\gamma}}{2}\Phi_a^{\alpha}V_p^{\nu}\partial_{\nu}\Phi_{\alpha b}
\end{aligned}$$

For $\hat{\Omega}_{abp}$:

$$\begin{aligned}
\hat{\Omega}_{abp} &= \frac{1}{2}(\hat{V}_a^{\hat{\mu}}\hat{V}_b^{\nu} - \hat{V}_b^{\hat{\mu}}\hat{V}_a^{\nu})\partial_{\nu}\hat{V}_{\hat{\mu}p} \\
&= 0
\end{aligned}$$

The same holds for Ω_{abc} . Using (??), we can then obtain the ω 's straightforwardly:

$$\hat{\omega}_{pmn} = \delta^{-\gamma}[\omega_{pmn} + \gamma(\eta_{pm}V_n^{\lambda} - \eta_{pn}V_m^{\lambda})\partial_{\lambda}], \quad (18a)$$

$$\hat{\omega}_{pma} = \kappa\delta^{-2\gamma}F_{pm}^{\alpha}\Phi_{\alpha a}, \quad (18b)$$

$$\hat{\omega}_{pab} = \frac{1}{2}\delta^{-\gamma}\Phi_a^{\alpha}V_p^{\mu}\partial_{\mu}\Phi_{\alpha\beta} - (a \leftrightarrow b), \quad (18c)$$

$$\hat{\omega}_{cmn} = -\kappa\delta^{-2\gamma}F_{mn}^{\alpha}\Phi_{\alpha c}, \quad (18d)$$

$$\hat{\omega}_{cab} = 0 \quad (18e)$$

We can now conclude this part by just computing the multiplication and look at what we got. This gives us.

$$\boxed{
\begin{aligned}
S &= \int dx^D \int V \delta\left(\frac{-1}{4}g^{\rho\lambda}\partial_{\lambda}h_{\alpha\beta}\partial_{\rho}h^{\alpha\beta} + \frac{1}{4}F^{\mu\nu\alpha}F_{\mu\nu}^{\beta}h_{\alpha\beta} + \right. \\
&\quad \left. + R + \frac{1}{4}\partial_{\hat{\mu}}\log(h)\partial^{\hat{\mu}}\log(h) + \partial_{\hat{\mu}}\log(h)\partial^{\hat{\mu}}\hat{\phi}\right)
\end{aligned}
} \quad (19)$$

We can observe some interesting things. Firstly we can see that we have refound the classical Einstein action, coupled with E electromagnetism field. So what we can conclude is that gravity in D+E dimension is gravity in 4d plus E electromagnetism field. and we find a few other field which will come in handy for the dilaton field reduction.

4.5 Dilaton field reduction

For this part we will have to sum $S_{\hat{g}}$ and $S_{\hat{\phi}}$ that gave us

$$S_{\hat{g}+\hat{\phi}} = \int dx^D \int V \delta \left(\frac{-1}{4} g^{\rho\lambda} \partial_\lambda h_{\alpha\beta} \partial_\rho h^{\alpha\beta} + \frac{1}{4} F^{\mu\nu\alpha} F_{\mu\nu}^\beta h_{\alpha\beta} + \right. \\ \left. + R + \frac{1}{4} \partial_{\hat{\mu}} \log(h) \partial^{\hat{\mu}} \log(h) + \partial_{\hat{\mu}} \log(h) \partial^{\hat{\mu}} \hat{\phi} + \partial_{\hat{\mu}} \hat{\phi} \partial^{\hat{\mu}} \hat{\phi} \right)$$

We then introduce the shifted dilaton as

$$\phi = \hat{\phi} - \frac{1}{2} \log h = \hat{\phi} - \log \delta \quad (20)$$

wich gave us

$$S = \int dx^D V e^{-\phi} \left(R - \frac{1}{4} F^{\mu\nu\alpha} F_{\mu\nu}^\beta h_{\alpha\beta} + \right. \\ \left. - g^{\rho\lambda} \partial_\lambda h_{\alpha\beta} \partial_\rho h^{\alpha\beta} + \hat{g}^{\hat{\mu}\hat{\nu}} \partial_{\hat{\mu}} \phi \partial_{\hat{\nu}} \phi \right) \quad (21)$$

4.6 Two form fiel reduction

4.6.1 The reduction

For the two form we will reduce it apart from the two other field, this those not cause any problem as the action is in effect linear.

We can rewrite (??) as

$$\begin{aligned} \mathcal{L}_{\hat{B}} &= \frac{-1}{12} (g^{\hat{\mu}\hat{\mu}'} g^{\hat{\nu}\hat{\nu}'} g^{\hat{\rho}\hat{\rho}'} \hat{H}_{\hat{\mu}\hat{\nu}\hat{\rho}} \hat{H}_{\hat{\mu}'\hat{\nu}'\hat{\rho}'}) \\ &= \frac{-1}{12} (\hat{\eta}^{\hat{r}\hat{r}'} \hat{\eta}^{\hat{s}\hat{s}'} \hat{\eta}^{\hat{t}\hat{t}'} \hat{H}_{\hat{r}\hat{s}\hat{t}} \hat{H}_{\hat{r}'\hat{s}'\hat{t}'}) \\ &= \frac{-1}{12} (\hat{H}_{\hat{r}\hat{s}\hat{t}} \hat{H}^{\hat{r}\hat{s}\hat{t}}) \\ &= \frac{-1}{12} (\hat{H}_{rst} \hat{H}^{rst} + \hat{H}_{rsc} \hat{H}^{rsc} + \hat{H}_{rbt} \hat{H}^{rbt} + \hat{H}_{rbc} \hat{H}^{rbc} + \hat{H}_{ast} \hat{H}^{ast} + \hat{H}_{asc} \hat{H}^{asc} + \hat{H}_{abs} \hat{H}^{abs} + \hat{H}_{abc} \hat{H}^{abc}) \end{aligned}$$

however thanks to to the fact that $\hat{B}_{\hat{\mu}\hat{\nu}}$ is antysymmetric one can easily show that this imply that $\hat{H}_{\hat{\mu}\hat{\nu}\hat{\rho}}$ is antysemtic for any neighbor index. so we obtaint

$$\mathcal{L}_{\hat{B}} = \frac{-1}{12} (\hat{H}_{rst} \hat{H}^{rst} + 3 \hat{H}_{rsc} \hat{H}^{rsc} + 3 \hat{H}_{asc} \hat{H}^{asc} + \hat{H}_{abc} \hat{H}^{abc}) \quad (22)$$

and finnaly the action of B is writen as

$$S_{\hat{b}} = - \int dx^D \sqrt{-g} e^{-\phi} \left\{ \frac{1}{12} H_{\alpha\beta\gamma} H^{\alpha\beta\gamma st} + \frac{1}{4} H_{\mu\alpha\beta} H^{\mu\alpha\beta} + \frac{1}{4} H_{\mu\nu\alpha} H^{\mu\nu\alpha} + \frac{1}{12} H_{\mu\nu\rho} H^{\mu\nu\rho} \right\} \quad (23)$$

The absence if hat is due to what we so in part 2

so with that we obtaint(23)

Now we will compute each componant of non hated H in function of hated H

to begin slowly as we have seen above $\hat{B}_{\alpha\beta} = B_{\alpha\beta}$ and if we use the y independence we get $H_{\alpha\beta\gamma} = 0$

We continue with $H_{\mu\alpha\beta}$

$$\begin{aligned} H_{\mu\alpha\beta} &= V_\mu^r \hat{V}_r^{\hat{\mu}} \hat{H}_{\hat{\mu}\alpha\beta} \\ &= \partial_\mu B_{\alpha\beta} \end{aligned} \quad (24)$$

those two were quite easy howver the next two are going to be more troublesome, but we are going to introduce new object Now for $H_{\mu\nu\beta}$

$$\begin{aligned} H_{\mu\alpha\beta} &= V_\mu^r V_\nu^s \hat{V}_r^{\hat{\mu}} \hat{V}_s^{\hat{\nu}} \hat{H}_{\hat{\mu}\hat{\nu}\alpha} \\ &= \hat{H}_{\mu\nu\alpha} - (A_\mu^{(1)\alpha} \hat{H}_{\alpha\nu\rho} + A_\rho^{(1)\alpha} \hat{H}_{\mu\alpha\nu}) \\ &= \partial_\mu \hat{B}_{\nu\alpha} + \partial_\nu \hat{B}_{\alpha\mu} - A_\mu^{(1)\beta} \partial_\nu \hat{B}_{\alpha\beta} - A_\nu^{(1)\beta} \partial_\mu \hat{B}_{\beta\alpha} \\ &= \partial_\mu \hat{B}_{\nu\alpha} + \partial_\nu \hat{B}_{\alpha\mu} - \partial_\nu (A_\mu^{(1)\beta} \hat{B}_{\alpha\beta}) + \partial_\nu A_\mu^{(1)\beta} \hat{B}_{\alpha\beta} - \partial_\mu (A_\nu^{(1)\beta} \hat{B}_{\beta\alpha}) + \partial_\mu A_\nu^{(1)\beta} \hat{B}_{\beta\alpha} \\ &= \partial_\mu \hat{B}_{\nu\alpha} - \partial_\nu \hat{B}_{\mu\alpha} - \partial_\nu (A_\mu^{(1)\beta} \hat{B}_{\alpha\beta}) + \partial_\nu A_\mu^{(1)\beta} \hat{B}_{\alpha\beta} + \partial_\mu (A_\nu^{(1)\beta} \hat{B}_{\beta\alpha}) - \partial_\mu A_\nu^{(1)\beta} \hat{B}_{\beta\alpha} \\ &= \partial_\mu (\hat{B}_{\nu\alpha} + A_\nu^{(1)\beta} \hat{B}_{\alpha\beta}) - \partial_\nu (\hat{B}_{\mu\alpha} + A_\mu^{(1)\beta} \hat{B}_{\alpha\beta}) - \hat{B}_{\alpha\beta} (\partial_\mu A_\nu^{(1)\beta} - \partial_\nu A_\mu^{(1)\beta}) \\ &= F_{\mu\nu\alpha}^{(2)} - \hat{B}_{\alpha\beta} F_{\mu\nu}^{(1)\beta} \end{aligned} \quad (25)$$

and finally for $H_{\mu\nu\rho}$ we are just going to use the same technique as before which gave us

$$\begin{aligned} H_{\mu\nu\rho} &= V_\mu^r V_\nu^s V_\rho^t \hat{V}_r^{\hat{\mu}} \hat{V}_s^{\hat{\nu}} \hat{V}_t^{\hat{\rho}} \hat{H}_{\hat{\mu}\hat{\nu}\hat{\rho}} \\ &= \hat{H}_{\mu\nu\rho} - A_\mu^{(1)\alpha} \hat{H}_{\alpha\nu\rho} - A_\nu^{(1)\alpha} \hat{H}_{\alpha\rho\mu} - A_\rho^{(1)\alpha} \hat{H}_{\alpha\mu\nu} + A_\mu^{(1)\alpha} A_\nu^{(1)\beta} \hat{H}_{\alpha\beta\rho} + A_\nu^{(1)\alpha} A_\rho^{(1)\beta} \hat{H}_{\alpha\beta\mu} + A_\rho^{(1)\alpha} A_\mu^{(1)\beta} \hat{H}_{\alpha\beta\nu} \\ &= \partial_\mu B_{\nu\rho} - \frac{1}{2}(A_\mu^{(1)\alpha} F_{\nu\rho\alpha}^{(2)} + A_{\mu\alpha}^{(2)} F_{\nu\rho}^{(1)}) + \partial_\nu B_{\rho\mu} - \frac{1}{2}(A_\nu^{(1)\alpha} F_{\rho\mu\alpha}^{(2)} + A_{\nu\alpha}^{(2)} F_{\rho\mu}^{(1)}) + \partial_\rho B_{\mu\nu} - \frac{1}{2}(A_\rho^{(1)\alpha} F_{\mu\nu\alpha}^{(2)} + A_{\rho\alpha}^{(2)} F_{\mu\nu}^{(1)}) \end{aligned} \quad (26)$$

where

$$B_{\mu\nu} = \hat{B}_{\mu\nu} + \frac{1}{2}A_\mu^{(1)\alpha} A_\nu^{(2)} - \frac{1}{2}A_\nu^{(1)\alpha} A_\mu^{(2)} - A_\mu^{(1)\alpha} B_{\alpha\beta} A_\nu^{(2)\beta} \quad (27)$$

4.6.2 Observation of gauge

Now as in part ?? we are going to observe the gauge rule to be sure that every object transforms as it should so let's begin with A^2 which should transform with the Lie derivative and under the new transformations of $B_{\mu\nu} \rightarrow B_{\mu\nu} + \partial_\mu u \Lambda_\nu$, so let's begin with $A_{\mu\alpha}^{(2)}$

$$\begin{aligned} \delta A_{\mu\alpha}^{(2)} &= \delta(\hat{B}_{\mu\alpha} + B_{\alpha\beta} A_\mu^{(1)\beta}) \\ &= \delta(\hat{B}_{\mu\alpha}) + \delta(B_{\alpha\beta} A_\mu^{(1)\beta}) \\ &= \delta(\hat{B}_{\mu\alpha}) + \delta(B_{\alpha\beta}) A_\mu^{(1)\beta} + B_{\alpha\beta} \delta(A_\mu^{(1)\beta}) \\ &= L_\xi \hat{B}_{\mu\alpha} + \partial_\mu \Lambda_\alpha^{(2)} + A_\mu^{(\beta)} L_\xi \hat{B}_{\alpha\beta} + \hat{B}_{\alpha\beta} \partial_\mu \Lambda_\alpha^{(1)} \\ &= L_\xi \hat{B}_{\mu\alpha} + \partial_\mu \Lambda_\alpha^{(2)} + L_\xi(A_\mu^{(\beta)} \hat{B}_{\alpha\beta}) \\ &= \partial_\mu \Lambda_\alpha^{(2)} + L_\xi A_{\mu\alpha}^{(2)} \end{aligned} \quad (28)$$

so we have the right transformation for this object

Doing the same thing with $B_{\mu\nu}$ we come under the transformation

$$\delta B_{\mu\nu} = \frac{1}{2}(\Lambda^{(1)\alpha} F_{\mu\nu\alpha}^{(2)} + \lambda_\alpha^{(2)} F_{\mu\nu}^{(1)\alpha}) \quad (29)$$

which is exactly the good term to compensate the additional term for the transformation of $H_{\mu\nu\rho}$

4.7 Complete reduced action

To recap we have the complete reduced action in the form

$$S = \int dx^D \sqrt{(-g)} e^{-\phi} \mathcal{L} \quad (30)$$

Where we decompose $\mathcal{L}$ as $\mathcal{L} = \mathcal{L}_1 + \mathcal{L}_2 + \mathcal{L}_3 + \mathcal{L}_4$, and we pose

$$\begin{aligned} \mathcal{L}_1 &= R + g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi \\ \mathcal{L}_2 &= \frac{1}{4} g^{\mu\nu} (\partial_\mu h_{\alpha\beta} \partial_\nu G^{\alpha\beta} - h^{\alpha\beta} h^{\gamma\delta} \partial_\mu B_{\alpha\gamma} \partial_\nu B_{\beta\delta}) \\ \mathcal{L}_3 &= -\frac{1}{4} g^{\mu\rho} g^{\nu\lambda} (h_{\alpha\beta} F_{\mu\nu}^{(1)\alpha} F_{\rho\lambda}^{(1)\beta} + h^{\alpha\beta} H_{\mu\nu\alpha} H_{\rho\lambda\beta}) \\ \mathcal{L}_4 &= -\frac{1}{12} H_{\mu\nu\rho} H^{\mu\nu\rho} \end{aligned} \quad (31)$$

5 symmetries observation

Now for this part we are going to explore a very important part of physics research nowadays which is symmetry. In our context it will allow us to make equivalence between multiple theories as a whole. But firstly we have to see the obvious symmetry that we have seen in the non-reduced version of the Action which is the $SL(E, R)$. But what will interest us in the reduction is that it does not create nor destroy any symmetry and if we observe a new symmetry in the reduced form that means that we have effectively discovered a new symmetry in the unreduced form. and following the [5] we observe an $O(E, E)$ symmetry. let's define it and then observe if it is true so the definition.

Definition 1 Any matrices M is in $O(E, E)$ if it is a square $E \times E$ matrices and if for the matrices

$$\zeta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

We can use this matrices as a metric for the group $O(E, E)$ as it has $d-1$ eigenvalues and d 1 eigenvalues, in a basis rotated from the one with diagonal metric.

the matrices M verify

$$M^T \zeta M = \zeta$$

We will now define a new matrices M which is

$$M = \begin{pmatrix} h^{-1} & -h^{-1}B \\ Bh^{-1} & h - Bh^{-1}B \end{pmatrix} \quad (32)$$

Of course we will verify that this true

$$\begin{aligned} M^T \zeta M &= \begin{pmatrix} h^{-1} & Bh^{-1} \\ -h^{-1}B & h - Bh^{-1}B \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} h^{-1} & -h^{-1}B \\ Bh^{-1} & h - Bh^{-1}B \end{pmatrix} \\ &= \begin{pmatrix} Bh^{-1} & h^{-1} \\ h - Bh^{-1}B & -h^{-1}B \end{pmatrix} \begin{pmatrix} h^{-1} & -h^{-1}B \\ Bh^{-1} & h - Bh^{-1}B \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \zeta \end{aligned}$$

so we confirm that $M \in O(E, E)$ Now that we have that in mind let's retok (31) and verify that each $\mathcal{L}$ is infact $O(E, E)$ invariant

For $\mathcal{L}_1$ the invzriance is obvious as $g_{\mu\nu}$ and Φ are both invariant. Nonwithsandaning we ought to remamber that for Φ each terms in it are note on there self invariant only.

Now the [5] propose a new tactics to observe the invariance of $\mathcal{L}_2$ which is to rewrite everything in matricial terms. we the obtain

$$\mathcal{L}_2 = \frac{1}{4} \text{tr}(\partial_\mu h^{-1} \partial^\mu h + h^{-1} \partial_\mu B h^{-1} \partial^\mu B)$$

And now we are going to do a few manipulation

$$\begin{aligned} \mathcal{L}_2 &= \frac{1}{4} \text{tr}(\partial_\mu h^{-1} \partial^\mu h + h^{-1} \partial_\mu B h^{-1} \partial^\mu B) \\ &= \frac{1}{8} \text{tr}(2\partial_\mu h \partial^\mu h^{-1} + 2h^{-1} \partial_\mu B h^{-1} \partial^\mu B) \\ &= \frac{1}{8} \text{tr}(2\partial_\mu h \partial^\mu h^{-1} - 2\partial_\mu (Bh^{-1}B) \partial^\mu h^{-1} + \partial_\mu (Bh^{-1}) \partial^\mu (Bh^{-1}) + \partial_\mu (h^{-1}B) \partial^\mu (h^{-1}B)) \\ &= \frac{1}{8} \text{tr}(\partial_\mu (h - Bh^{-1}B) \partial^\mu h^{-1} + \partial_\mu (Bh^{-1}) \partial^\mu (Bh^{-1}) + \partial_\mu (h^{-1}B) \partial^\mu (h^{-1}B) + \partial^\mu h^{-1} \partial_\mu (h - Bh^{-1}B)) \\ &= \frac{1}{8} \text{tr}(\partial_\mu M^{-1} \partial^\mu M) \end{aligned} \quad (33)$$

Now we are going to rewrite $\mathcal{L}_3$ by developing the H partial to get

$$\begin{aligned} \mathcal{L}_3 &= -\frac{1}{4} [F_{\mu\nu}^{(1)\alpha} h_{\alpha\beta} F^{(1)\mu\nu\beta} + (F_{\mu\nu}^{(2)} - B_{\alpha\gamma} F_{\mu\nu}^{(1)\gamma}) h^{\alpha\beta} (F_{\beta}^{(2)\mu\nu} - B_{\beta\delta} F^{(1)\mu\nu\delta})] \\ &= -\frac{1}{4} \mathcal{F}_{\mu\nu}^i (M^{-1})_{ij} \mathcal{F}^{\mu\nu j} \end{aligned} \quad (34)$$

Where $\mathcal{F}_{\mu\nu}^i$ is the 2E-component vector of field strengths

$$\mathcal{F}_{\mu\nu}^i = \begin{pmatrix} F_{\mu\nu}^{(1)\alpha} \\ F_{\mu\nu}^{(2)} \end{pmatrix} = \partial_\mu \mathcal{A}_\nu^i - \partial_\nu \mathcal{A}_\mu^i \quad (35)$$

Now the $O(E, E)$ invariance is visible if we consider that the vector fields transform linealy to the vector representation of $O(E, E)$, i.e., $\mathcal{A}_\mu^i \rightarrow \Omega_j^i \mathcal{A}_\mu^j$ And finnaly we go to $\mathcal{L}_4$ firstly let's rewrite $H_{\mu\nu\rho}$

$$\begin{aligned} H_{\mu\nu\rho} &= \partial_\mu B_{\nu\rho} - \frac{1}{2} (A_\mu^{(1)\alpha} F_{\nu\rho\alpha}^{(2)} + A_{\mu\alpha}^{(2)} F_{\nu\rho}^{(1)}) + \partial_\nu B_{\rho\mu} - \frac{1}{2} (A_\nu^{(1)\alpha} F_{\rho\mu\alpha}^{(2)} + A_{\nu\alpha}^{(2)} F_{\rho\mu}^{(1)}) + \partial_\rho B_{\mu\nu} - \frac{1}{2} (A_\rho^{(1)\alpha} F_{\mu\nu\alpha}^{(2)} + A_{\rho\alpha}^{(2)} F_{\mu\nu}^{(1)}) \\ &= \partial_\mu B_{\nu\rho} - \frac{1}{2} \mathcal{A}_\mu^i \zeta_{ij} \mathcal{F}_{\nu\rho}^j + (cyc.perms) \end{aligned} \quad (36)$$

Now as $\Omega^T \zeta \Omega = \zeta$ the seconds thermes is $O(E, E)$ invariant. And the for the first terme is invariant if we require that $B_{\nu\rho}$ does not transfrom. This demonstrates that the action hides a non-trivial symmetry, providing a useful framework for comparing equivalent theories.

6 Conclusion

References

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A lagrangian

By using the principle of least action and the variational principle we can see that

$$\begin{aligned}
 \delta S &= \delta \int \mathcal{L} dx^\mu \\
 &= \delta \int \frac{\partial \mathcal{L}}{\partial \eta_\rho} \delta \eta_\rho + \frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho} \delta \partial_\nu \eta_\rho dx^\mu \\
 &= \delta \int \frac{\partial \mathcal{L}}{\partial \eta_\rho} \delta \eta_\rho - \partial_\nu \frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho} \delta \eta_\rho + \underbrace{\partial_\nu \left(\frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho} \delta \eta_\rho \right)}_{\text{border terme equal to zero}} dx^\mu \\
 &= 0
 \end{aligned}$$

which gave the equation

$$\boxed{\frac{\partial \mathcal{L}}{\partial \eta_\rho} - \partial_\nu \frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho} = 0} \quad (37)$$

which is in fact the equation of mouvement for the field

B General relativty 101

Definition 2 Let V be an R -vector space of dimension N , we define the tensor of order $(r, s) \in \llbracket 0, 1 \rrbracket^2$ on $V^{(r,s)} \stackrel{\text{def}}{=} V^{\otimes r} \otimes V^{\otimes s}$ as a being a multilinear application on $V^{*r} \times V^s$ to $\mathbb{R}$

Definition 3 We will call n -dimension ($n \in \mathbb{N}^*$) manifold every Hausdorff topological space $(\mathcal{M}, \tau)$ locally homeomorphic to $\mathbb{R}^n$)

Definition 4 Let $(\mathcal{M}, g)$ be a smooth manifold. A metric tensor is a symmetric, non-degenerate bilinear form $g_{\mu\nu}$ on the tangent space $T_p \mathcal{M}$ at each point $p \in \mathcal{M}$, such that

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu$$

In General Relativity we use the signature $(-, +, +, +)$.

Definition 5 In general in RG we use the christoffels connexion as it is the sole one wich is mtreci and torsion free. We define it as being

$$\text{nabla}_\mu V^\nu = \partial_\mu V^\nu + \Gamma_{\mu\sigma}^\nu V^\sigma \quad (38)$$

where the christoffels symbols are

$$\Gamma_{\mu\nu}^\lambda = \frac{1}{2} g^{\lambda\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}) \quad (39)$$